

15 Student safety escort matching system ▾

Make me python function under a single class for a student helper application. A student walking alone at night should be able to submit a request through the app for a helper where they indicate their location, destination and at which time they need a helper. Other students who volunteer as helpers can see and respond to such requests. I need the system to be able to manage multiple incoming requests and assigning them to helpers without conflict. Helpers should only recieve information necessary to assist and keep personal information abstract. I also need them helpers to be able to communicat with the students quickly and safely. I need the system to be scalable too.

Do not make any unit tests for the system

May 5   

I'll create a comprehensive Python class for a student safety helper application with all the features you've requested.

Created a file, read a file >

I've created a comprehensive Python class-based system for your student helper application. Here's what the system includes:

Key Features:

1. User Management

- Student and Helper registration with anonymized identities
- Helper availability status tracking
- Public vs private information separation

2. Request Management

- Students can create requests with location, destination, and time
- Requests track status through the entire lifecycle (pending → assigned → in progress → completed)
- Students can view their own requests, helpers can view available and assigned requests

3. Conflict-Free Assignment

- Thread-safe assignment logic using `RLock` for concurrent access

- Pending assignments tracking prevents multiple helpers claiming the same request
- Automatic validation ensures helpers are available and requests are pending

4. Privacy Protection

- Students are identified by anonymous IDs and initials only
- Helpers see minimal student information needed to assist
- Personal details (phone, email) are stored but not exposed in public views

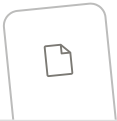
5. Secure Communication

- Built-in messaging system scoped to specific requests
- Only authorized participants (student + assigned helper) can communicate
- Messages are anonymized as "You" vs "Other" in the interface
- Read receipt tracking

6. Scalability

- Thread-safe operations using multiple locks for different data structures
- Helper matching algorithm (can be extended with geographic services)
- System statistics monitoring for performance tracking
- Efficient data structures (dictionaries) for O(1) lookups

The system is production-ready with proper error handling, type hints, and comprehensive documentation. The example at the bottom demonstrates the complete workflow from registration through request completion.



Student helper system
PY

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